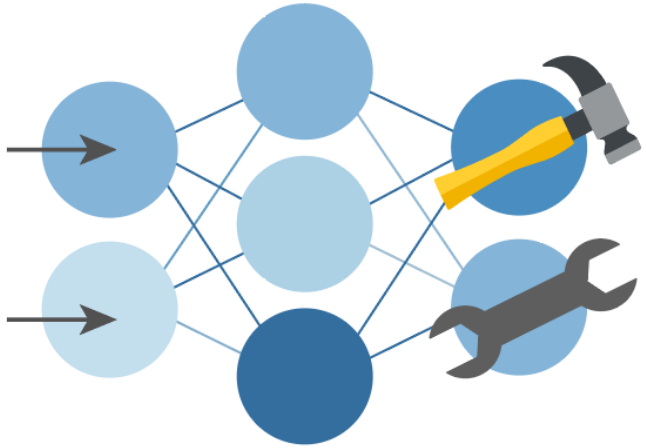
Python + Agents



🛠️ Feb 24: Building your first agent in Python
🧠 Feb 25: Adding context and memory to agents
📊 Feb 26: Monitoring and evaluating agents
🔀 Mar 3: Building your first AI-driven workflows
🕸️ Mar 4: Orchestrating advanced multi-agent workflows
🙋🏽‍♀️ Mar 5: Adding a human-in-the-loop to workflows

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Microsoft

Python + Agents

🙋🏽‍♀️ Adding a human-in-the loop to workflows

aka.ms/pythonagents/slides/hitl

Pamela Fox

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Today we'll cover...

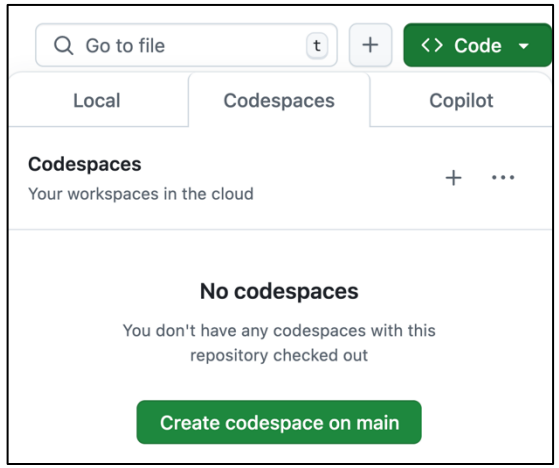
- Why HITL matters for agents and workflows
- Tool approval: gating sensitive operations
- Requests and responses: structured human interaction
- Checkpoints and resuming: durable HITL workflows
- Handoff with HITL: interactive multi-agent routing
- E2E application with workflows

Want to follow along?

1. Open this GitHub repository:

aka.ms/python-agentframework-demos

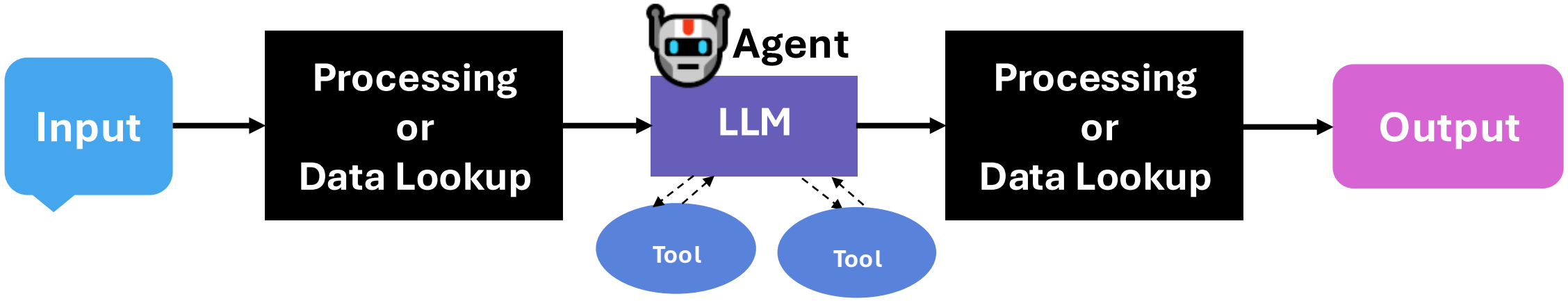
2. Use "Code" button to create a GitHub Codespace:



3. Wait a few minutes for Codespace to start up ⏱️
4. Run `git pull` to ensure you have the latest changes

Recap: What's an agentic workflow?

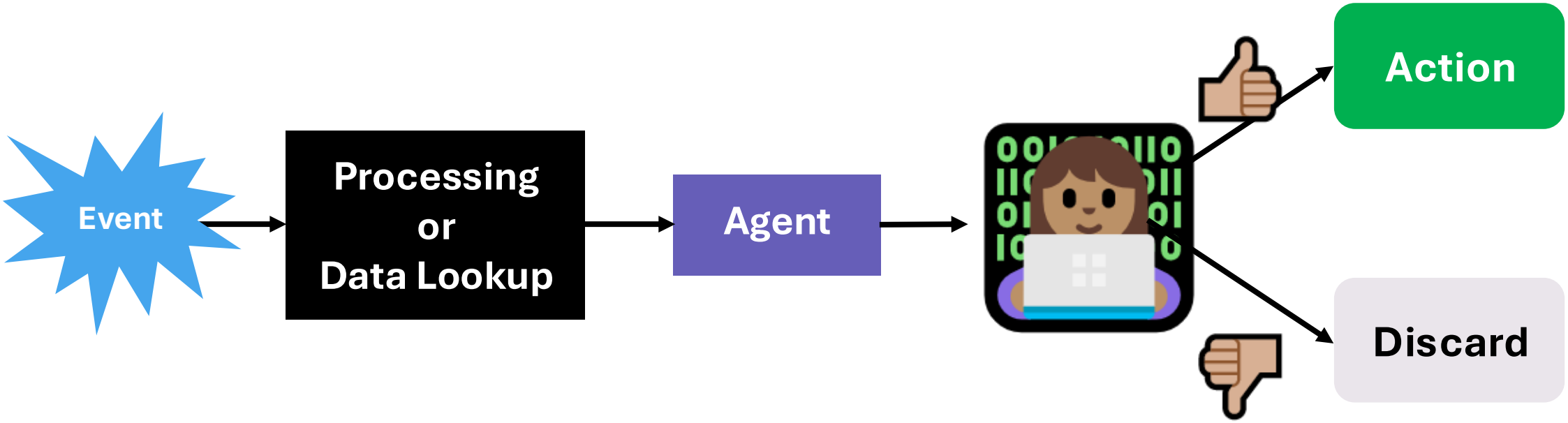
An agentic workflow is a flow that involves an agent at some point, typically to handle decision making or answer synthesis.



In agent-framework, a workflow is a graph with `Executor` nodes and edges between:

Human-in-the-loop

If an agent wants to take an action, a human should often review it first.



Why add a human in the loop?

Why add a human in the loop?

LLMs can produce uncertain, inconsistent, or incorrect outputs.

Human checkpoints provide:

🎯 **Accuracy**	Verify factual correctness before acting
🛡️ **Safety**	Gate sensitive operations (refunds, emails, deployments)
🤝 **Trust**	Users see what the system will do before it acts
📋 **Compliance**	Audit trail of human approvals for regulated workflows
🚦 **Control**	Humans can redirect, refine, or halt a workflow at any point

HITL patterns in agent-framework

Tool approval	Sensitive tool calls must be approved before execution
Requests and responses	Workflow needs human input to continue (answers, choices, feedback)
Checkpoints & resuming	Long-running tasks where human may not be immediately available
Handoff with HITL	Dynamic multi-agent routing with interactive user input between handoffs

Tool approval

https://learn.microsoft.com/agent-framework/agents/tools/tool-approval

When do we need tool approval?

Without approval:

- Agent decides to send email
- Tool executes automatically
- Email sent (possibly wrong!)

With approval:

- Agent wants to send email
- Workflow pauses
- Human reviews: "Send email?"
- Human approves ✓ or rejects ✗
- Tool executes only if approved

Consider requiring tool approval for:

- Financial operations (refunds, purchases)
- Communications (emails, messages)
- Destructive operations (deletions, deployments)
- Any irreversible action

Defining tools that require approval

Set `approval_mode` to either `"always_require"` or `"never_require"` (default):

```
@tool(approval_mode="never_require")
def lookup_receipt(
  receipt_id: Annotated[str, "The receipt ID to look up"],
) -> dict[str, str]:
  """Look up a receipt by ID and return its details."""
  ...

@tool(approval_mode="always_require")
def submit_expense_report(
  description: Annotated[str, "Description of the expense report"],
  total_amount: Annotated[str, "Total amount to reimburse"],
  receipt_ids: Annotated[str, "Comma-separated receipt IDs included"]
  ) -> str:
  """Submit an expense report for reimbursement. Requires approval."""
  ...
```

Full example: [agent_tool_approval.py]

Handling tool approval in the event loop

Check if the agent's result contains any `user_input_requests`:

```
result = await agent.run(query)

while len(result.user_input_requests) > 0:
  new_inputs: list[Any] = [query]

  for request in result.user_input_requests:
    func_call = request.function_call
    new_inputs.append(Message("assistant", [request]))
    approval = input(" Approve? (y/n): ").strip().lower()
    approved = approval == "y"
   new_inputs.append(Message("user",
     [request.to_function_approval_response(approved)]))

  result = await agent.run(new_inputs)
```

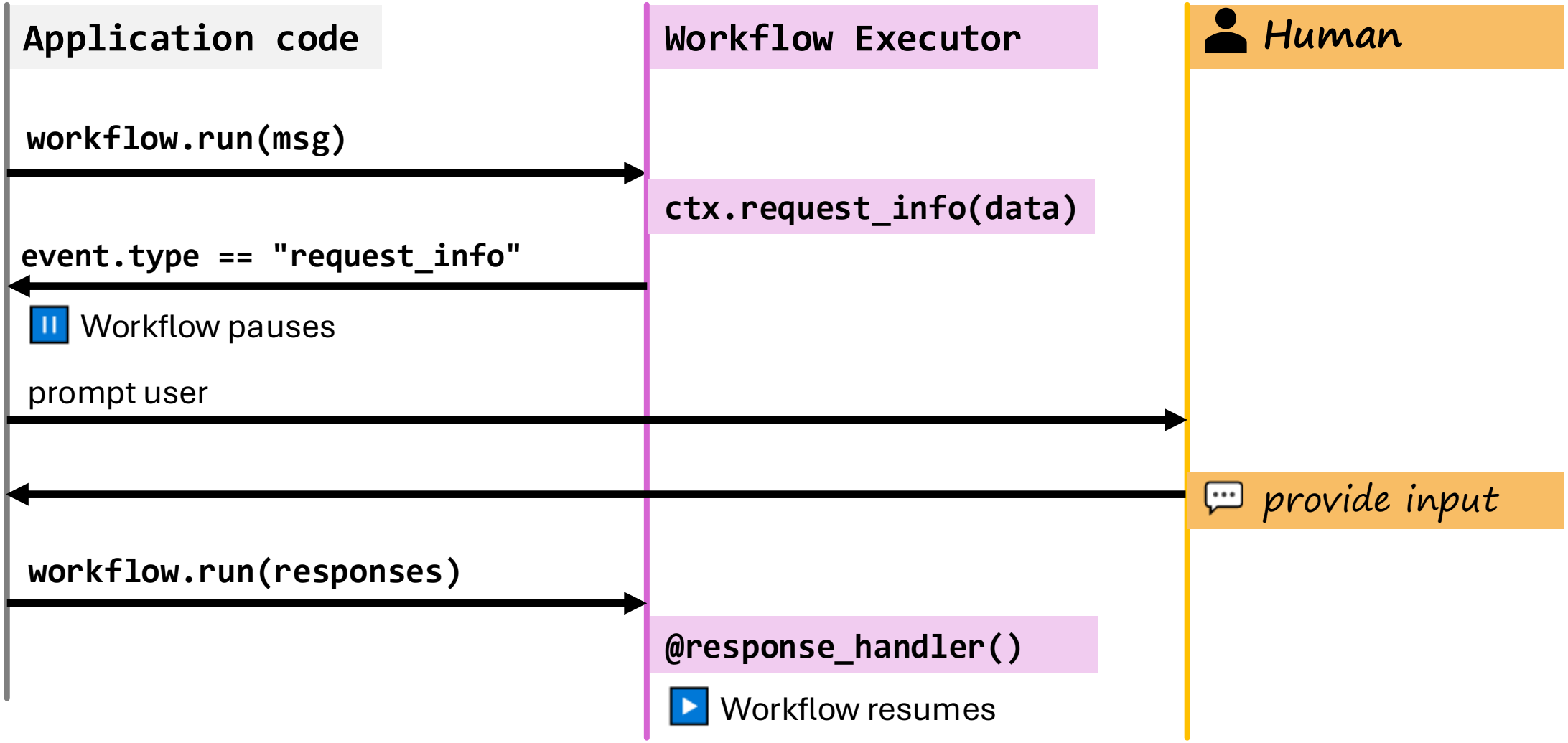
Full example: [agent_tool_approval.py]

Requests and responses

https://learn.microsoft.com/agent-framework/workflows/requests-and-responses

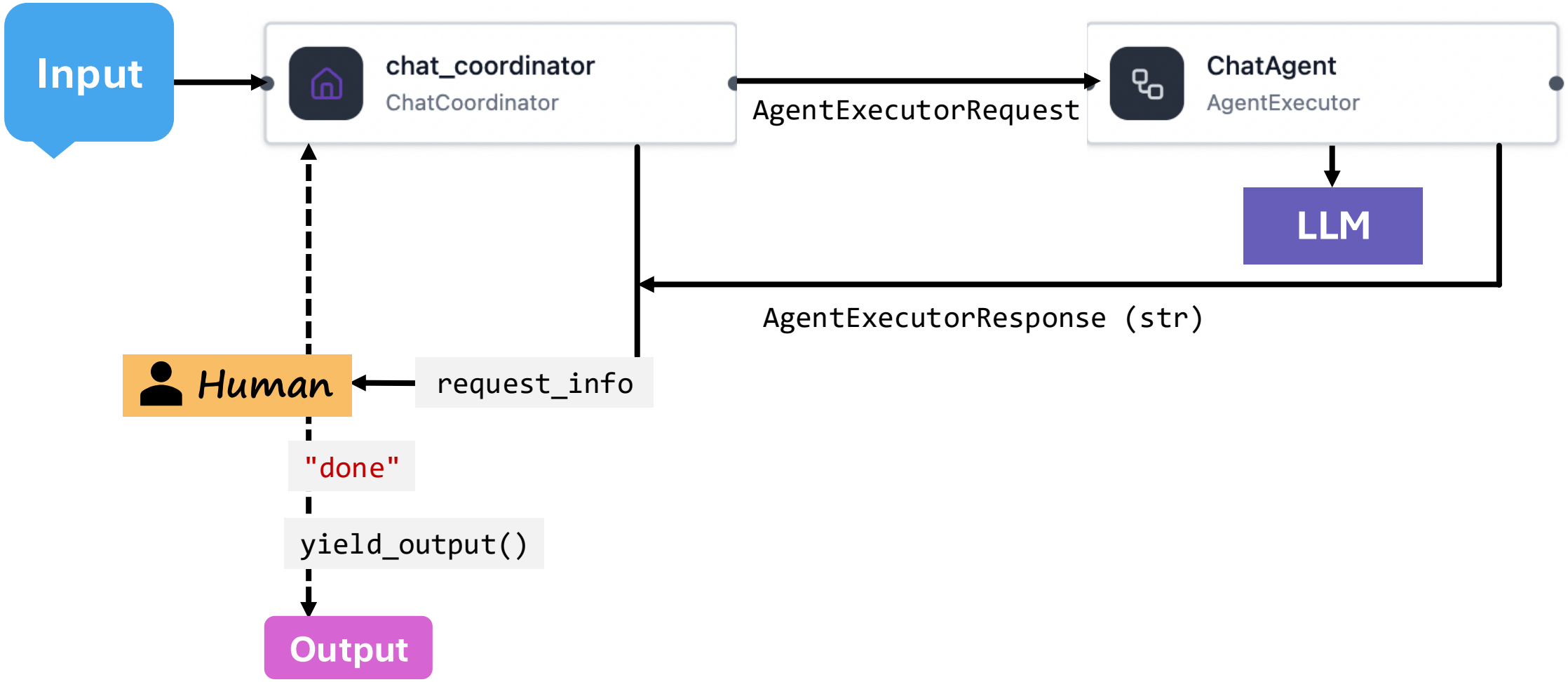
Requests and responses in workflows

Executors can pause a workflow and ask for human input:



Simple chat with HITL: Workflow topology

In this workflow, the coordinator calls `request_info` until user replies `"done"`:



Full example: [workflow_hitl_requests.py]

Executor code for handling requests and responses

Executor calls `request_info` to request HITL and defines `@response_handler` to handle reply:

```
class ChatCoordinator(Executor):

  @handler
  async def on_agent_response(self, result: AgentExecutorResponse,
                                  ctx: WorkflowContext) -> None:
    await ctx.request_info(
      request_data=UserPrompt(message=result.agent_response.text),
      response_type=str)

  @response_handler
  async def on_human_reply(self, original_request: UserPrompt, reply: str,
                       ctx: WorkflowContext[AgentExecutorRequest, str]) -> None:
    if reply.strip().lower() == "done":
      await ctx.yield_output("Conversation ended.")
      return
    await ctx.send_message(
       AgentExecutorRequest(messages=[Message("user", text=reply)],
                            should_respond=True), target_id=self._agent_id)
```

Full example: [workflow_hitl_requests.py]

App code for handling requests and responses

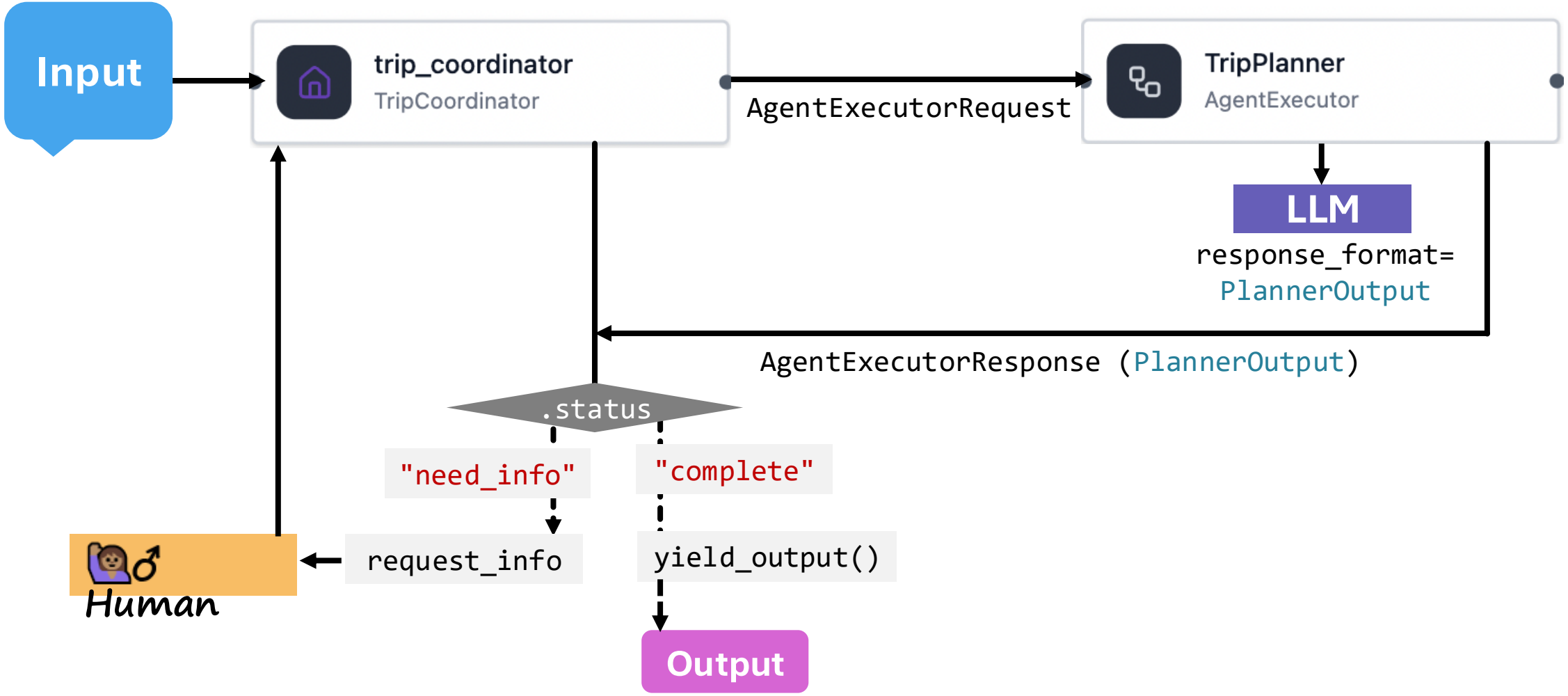
The app checks for `"request_info"` events, requests `input()`, runs workflow with responses:

```
stream = workflow.run(first_message, stream=True)

while True:
  pending = {}
  async for event in stream:
    if event.type == "request_info":
      pending[event.request_id] = event.data
    elif event.type == "output":
      print(f"\n{event.data}")
  if not pending:
    break

  for request_id, request in pending.items():
    print(f"\n🤖 Agent: {request.message}")
    reply = input("💬 You (or 'done'): ")
    pending[request_id] = reply

  stream = workflow.run(stream=True, responses=pending)
```

Full example: [workflow_hitl_requests.py]

HITL with structured outputs: Workflow topology

In this workflow, the coordinator calls `request_info` based on agent's structured output:



Full example: [workflow_hitl_requests_structured.py]

Agent definition with structured outputs

The agent must always respond with a `status` and either a `question` or `itinerary`:

```
class PlannerOutput(BaseModel):
  status: Literal["need_info", "complete"]
  question: str | None = None
  itinerary: str | None = None

planner_agent = Agent(
  name="TripPlanner",
  instructions="You are a helpful trip planner.",
  chat_client=client,
  default_options={"response_format": PlannerOutput},
)
```

Full example: workflow_hitl_requests_structured.py

Executor code for handling structured response

Executor calls `request_info` to request HITL based on the agent's structured response:

```
class TripCoordinator(Executor):
  @handler
  async def on_agent_response(self, result: AgentExecutorResponse,
                              ctx: WorkflowContext):
  output: PlannerOutput = result.agent_response.value
  if output.status == "need_info":
    await ctx.request_info(
       request_data=UserPrompt(message=output.question), response_type=str)
  else:
    await ctx.yield_output(output.itinerary)

  @response_handler
  async def on_human_answer(self, original_request: UserPrompt, answer: str,
                            ctx: WorkflowContext[AgentExecutorRequest, str]):
    user_msg = await ctx.send_message(
      AgentExecutorRequest(messages=[Message("user", text=answer)],
          should_respond=True), target_id=self._agent_id)
```

Full example: [workflow_hitl_requests_structured.py]

Checkpoints and resuming

https://learn.microsoft.com/agent-framework/workflows/checkpoints

The need for checkpoints in HITL workflows

Without checkpoints:

- Workflow pauses for human input
- Human is offline
- Process crashes or restarts
- All progress lost!

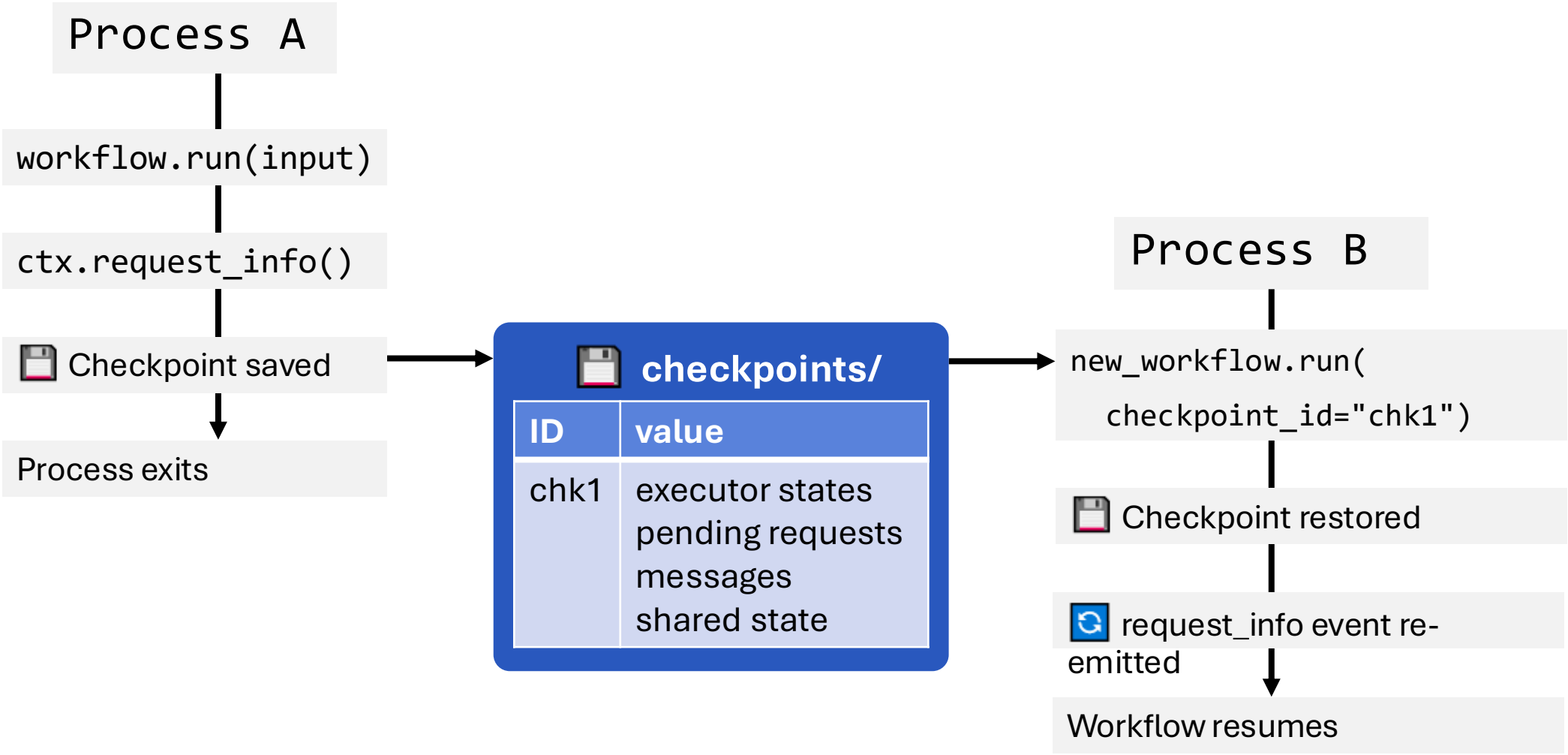
With checkpoints:

- Workflow pauses for human input
- Human is offline
- Checkpoint saved to disk 💾
- Process exits

...hours later...

- New process starts
- Loads checkpoint from disk 📂
- Human provides input
- Workflow resumes where it left off

Workflow lifecycle with checkpoints



Configuring checkpoint storage for workflows

Agent framework provides `InMemoryCheckpointStorage` or `FileCheckpointStorage`

Set `checkpoint_storage` to a built-in storage or a custom storage class:

```
from agent_framework import FileCheckpointStorage

storage = FileCheckpointStorage(storage_path=CHECKPOINT_DIR)

workflow = (WorkflowBuilder(
            start_executor=prepare_brief,
            checkpoint_storage=storage)
        .add_edge(prepare_brief, writer)
        .add_edge(writer, review_gateway)
        .add_edge(review_gateway, writer)
    .build())
```

Full example: workflow_hitl_checkpoint.py

Resuming a workflow from checkpoint

When starting the process, check if there is a saved checkpoint for the workflow. If so, pass the `checkpoint_id` in when running the workflow:

```
if checkpoint_id:
  stream = workflow.run(checkpoint_id=checkpoint_id, stream=True)
else:
  stream = workflow.run(brief, stream=True)

async for event in stream:
  if event.type == "request_info":
    pending[event.request_id] = event.data
  elif event.type == "output":
    print(f"\n✅ Workflow completed:\n{event.data}")
```

Full example: workflow_hitl_checkpoint.py

Your event processing loop does *not* change.

Any `"request_info"` events are re-emitted upon workflow restoration.

Resuming a workflow from checkpoint

Implement `on_checkpoint_save` and `on_checkpoint_restore` on Executors, to serialize any state specific to each instance.

```
class ReviewGateway(Executor):
  def __init__(self, id: str, writer_id: str) -> None:
    super().__init__(id=id)
    self._writer_id = writer_id
    self._iteration = 0

  async def on_checkpoint_save(self) -> dict[str, Any]:
    return {"iteration": self._iteration}

  async def on_checkpoint_restore(self, state: dict[str, Any]):
    self._iteration = state.get("iteration", 0)
```

Full example: [workflow_hitl_checkpoint.py]

Designing your custom checkpoint storage

To store the checkpoints in any backend, implement `CheckpointStorage` protocol:

Method	Purpose
`save(checkpoint)`	Persist a snapshot, return ID
`load(checkpoint_id)`	Restore a snapshot by ID
`delete(checkpoint_id)`	Remove a snapshot
`list_checkpoints(name)`	Retrieve all snapshots for a workflow
`list_checkpoint_ids(name)`	Retrieve only snapshot IDs for a workflow
`get_latest(name)`	Retrieve most recent snapshot for a workflow

Built-in backends:

- `FileCheckpointStorage`: JSON files on disk
- `InMemoryCheckpointStorage`: In-process dict

Custom backends:

Redis, PostgreSQL, Cosmos DB, …anything!

Custom checkpoint storage for workflows

To store the checkpoints in any backend, implement `CheckpointStorage` protocol:

id	workflow_name	timestamp	data
abc123	content_review	2026-03-03T00:38:15.893871	{"state": {"brief": "Introduce our new..
def456	content_review	2026-03-03T00:38:28.940753	{"state": {"brief": "Introduce our new..
ghi789	content_review	2026-03-03T00:38:29.207847	{"pending_request_info_events": ...

Design decisions to consider:

How to scope checkpoints?	Per workflow name? Per user/session? e.g. workflow_name = "review:{user_id}"
How long to keep checkpoints?	Prune after completion? TTL? Keep N latest?
How large are checkpoints?	Checkpoints include full message history. Long conversations = large JSONB blobs.
Who can access checkpoints?	Row-level security? Separate tables per tenant?

Custom checkpoint storage with PostgreSQL

Implement all the required methods of `CheckpointStorage` protocol:

```
class PostgresCheckpointStorage:

  async def save(self, checkpoint: WorkflowCheckpoint) -> str:
    encoded = encode_checkpoint_value(checkpoint.to_dict())
    async with await AsyncConnection.connect(self._conninfo) as conn:
      await conn.execute(
           """INSERT INTO workflow_checkpoints (id, workflow_name, timestamp, data)
                     VALUES (%s, %s, %s, %s)""",
           (checkpoint.checkpoint_id, checkpoint.workflow_name,
            checkpoint.timestamp, json.dumps(encoded)))
    return checkpoint.checkpoint_id

  async def load(self, checkpoint_id: str) -> WorkflowCheckpoint:
    async with await AsyncConnection.connect(self._conninfo) as conn:
      row = await (await conn.execute(
         "SELECT data FROM workflow_checkpoints WHERE id = %s", (checkpoint_id,),
         )).fetchone()
    return WorkflowCheckpoint.from_dict(decode_checkpoint_value(row["data"]))
```

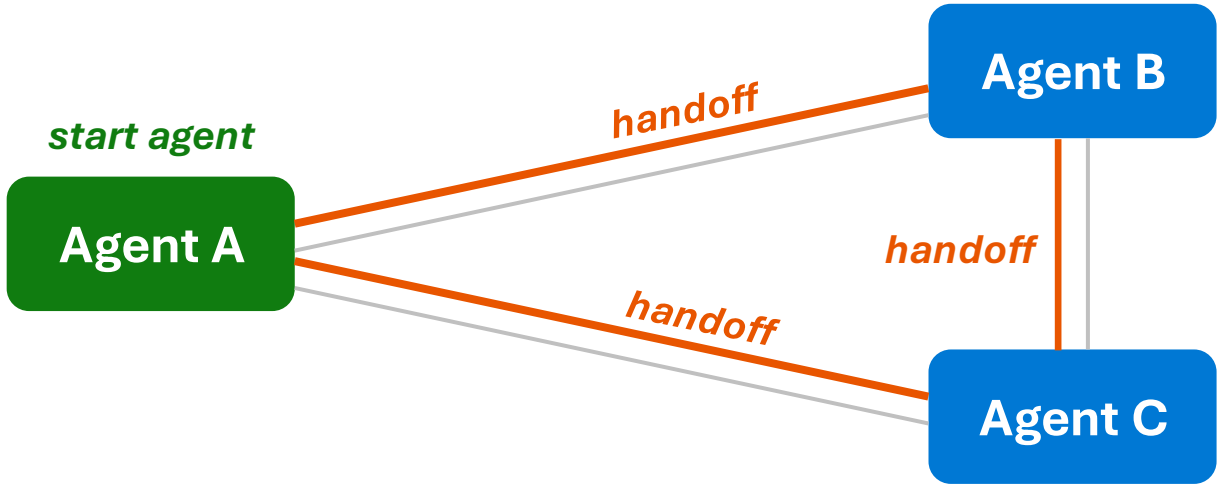
Full example: [workflow_hitl_checkpoint_pg.py]

Handoff orchestration with HITL

https://learn.microsoft.com/agent-framework/workflows/orchestrations/handoff

Review: Handoff orchestration

`HandoffBuilder` is a built-in workflow where agents can handoff to each other.

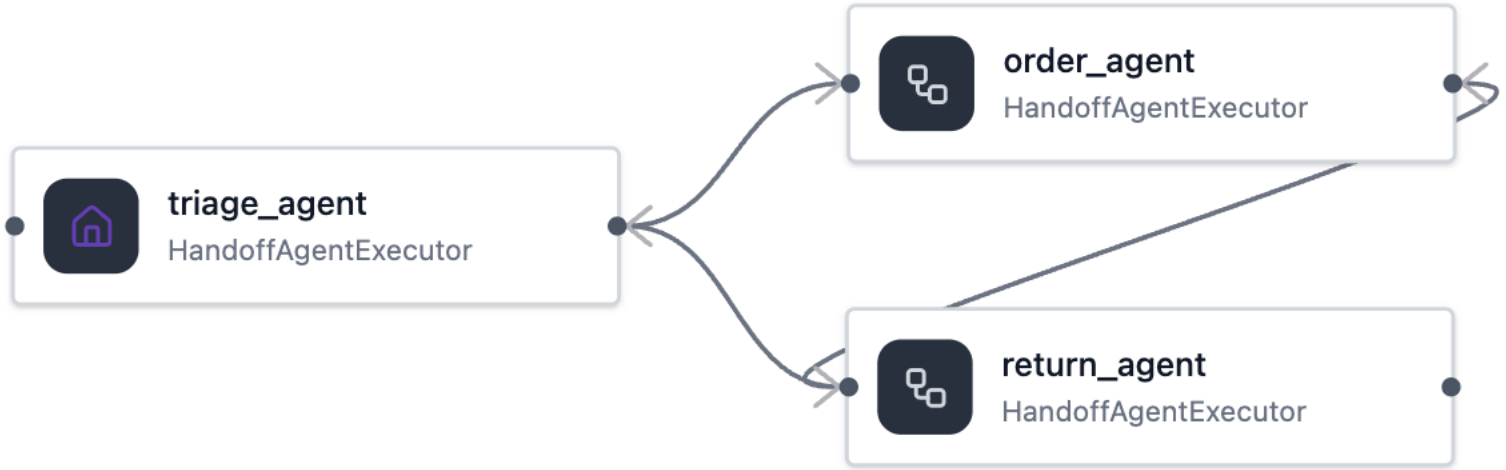


No edges are defined up front – routing emerges from conversation state.

Handoff is inherently interactive:

- When an agent doesn't handoff, it requests user input
- Workflow emits HandoffAgentUserRequest events
- Application must respond for the workflow to continue

Configuring HandoffBuilder for interactivity



`HandoffBuilder` is interactive as long as you don't set `with_autonomous_mode:`

```
workflow = (HandoffBuilder(
    name="customer_support",
    participants=[triage_agent, order_agent, return_agent],
     termination_condition=lambda conversation: (
     len(conversation) > 0 and "goodbye" in conversation[-1].text.lower()))
  .with_start_agent(triage_agent)
  .build())
```

Full example: [workflow_hitl_handoff.py]

Handling HITL requests in HandoffBuilder

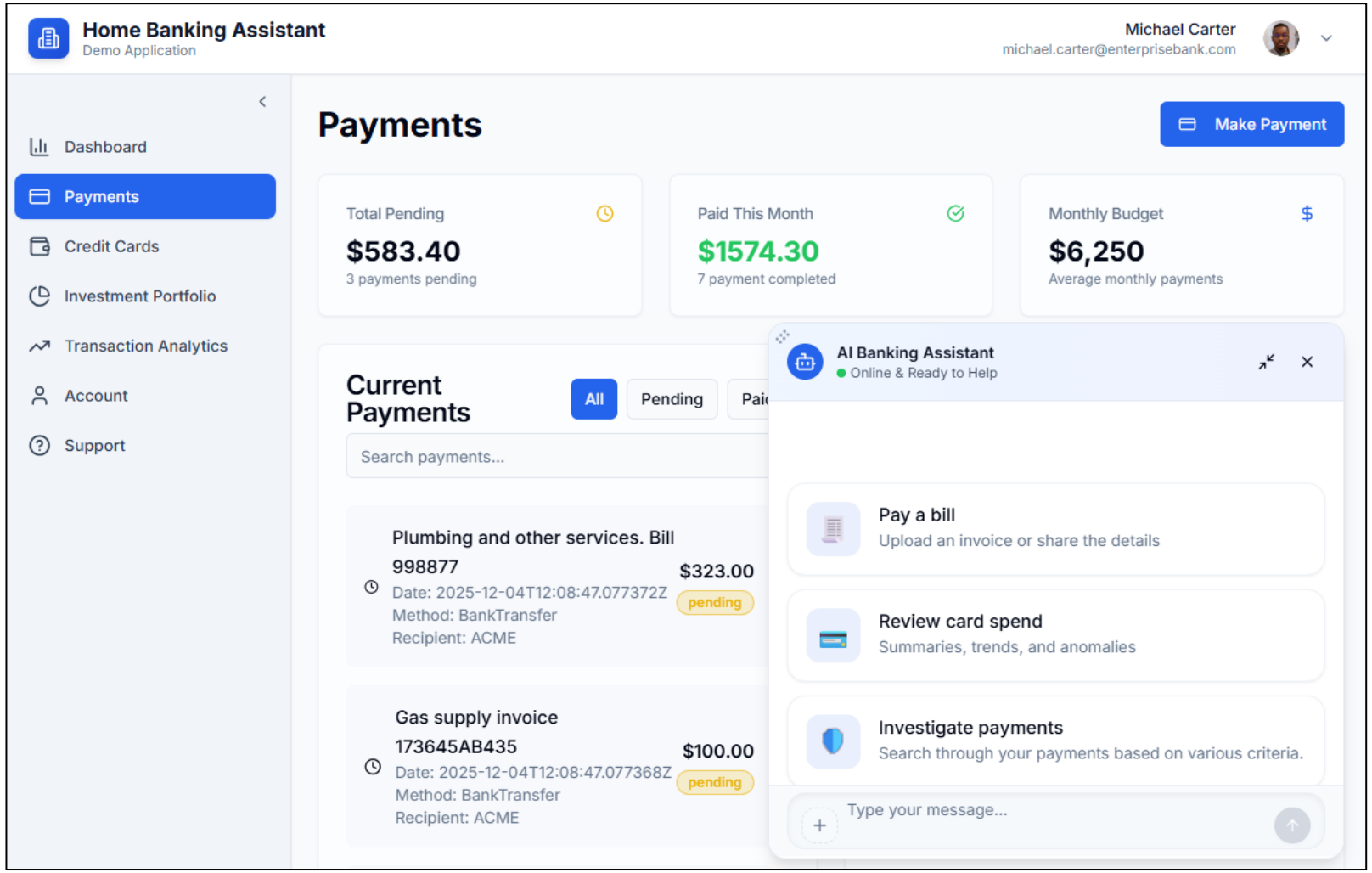
Handle all `HandoffAgentUserRequest` events and respond with user input:

```
responses: dict[str, Any] = {}
request_id = request_event.request_id

for request_event in pending:
  if isinstance(request_event.data, HandoffAgentUserRequest):
    agent_response = request_event.data.agent_response
    for msg in agent_response.messages:
      if msg.text:
        print(f"🤘 {msg.author_name}: {msg.text}")

user_input = input("\n🧑 You: ").strip()
if user_input.lower() in ("exit", "quit"):
  responses[request_id] = HandoffAgentUserRequest.terminate()
else:
  responses[request_id] = HandoffAgentUserRequest.create_response(user_input)

stream = workflow.run(responses=responses, stream=True)
```

Full example: [workflow_hitl_handoff.py]

End-to-end application

Agentic banking assistant

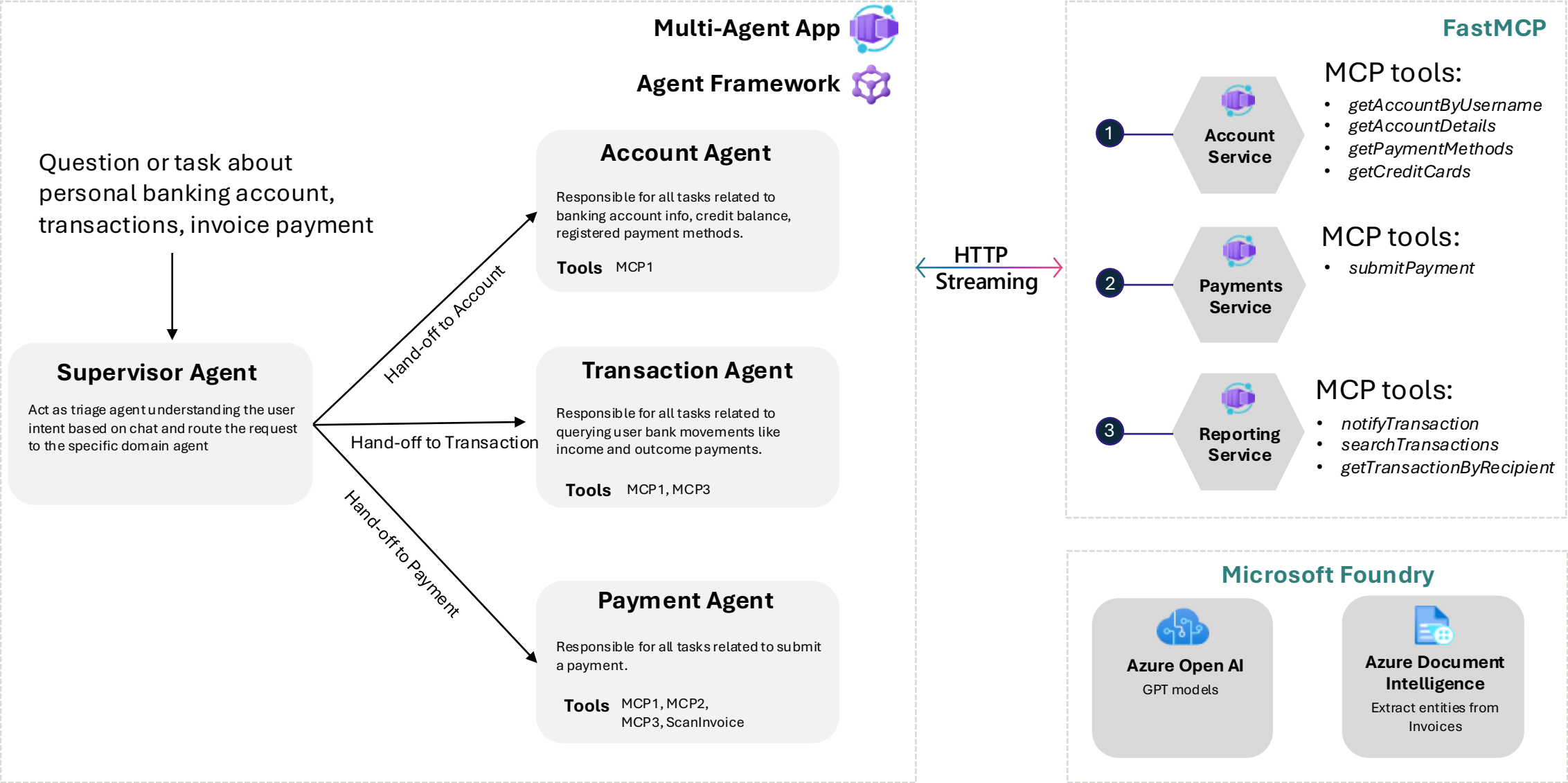


Python frameworks:
Agent Framework
FastMCP

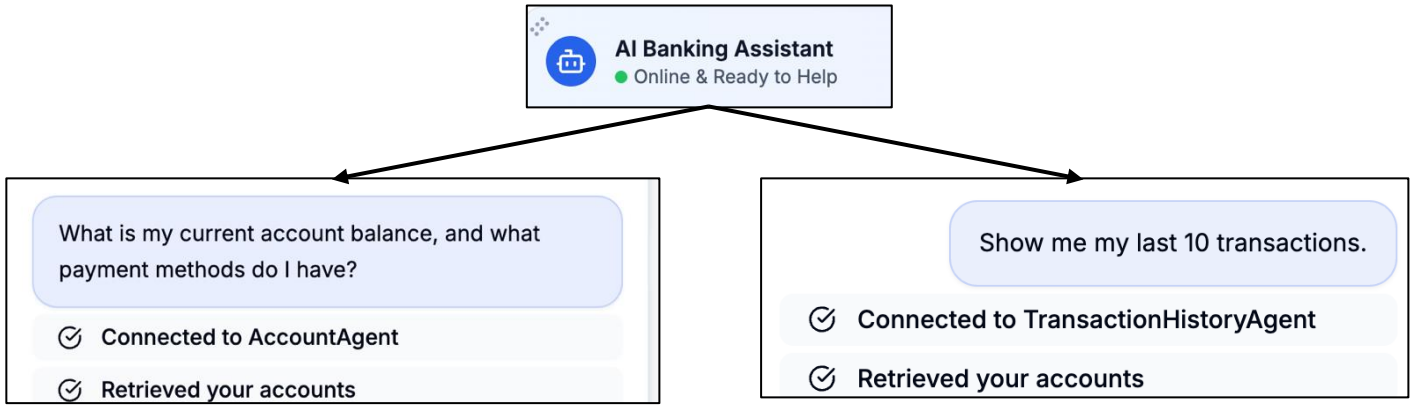
Azure services:
OpenAI
Document Intelligence
Container Apps

https://github.com/Azure-Samples/agent-openai-python-banking-assistant

Agents and MCP servers architecture



Handoff workflow with checkpointing




```
workflow = (HandoffBuilder(
    participants=[triage_agent,
    await self.account_agent.build_af_agent(),
    await self.transaction_agent.build_af_agent(),
    await self.payment_agent.build_af_agent()])
   .with_start_agent(triage_agent)
   .with_checkpointing(checkpoint_storage)
   .build())
```

[Full file: handoff_orchestrator.py]

Next steps

Watch past recordings:
aka.ms/pythonagents/resources

Join office hours after each session in Discord:
aka.ms/pythonai/oh

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